Explainable AI for Academic Integrity Verification

Progress since Meeting 4, and the run-in to the graded presentation

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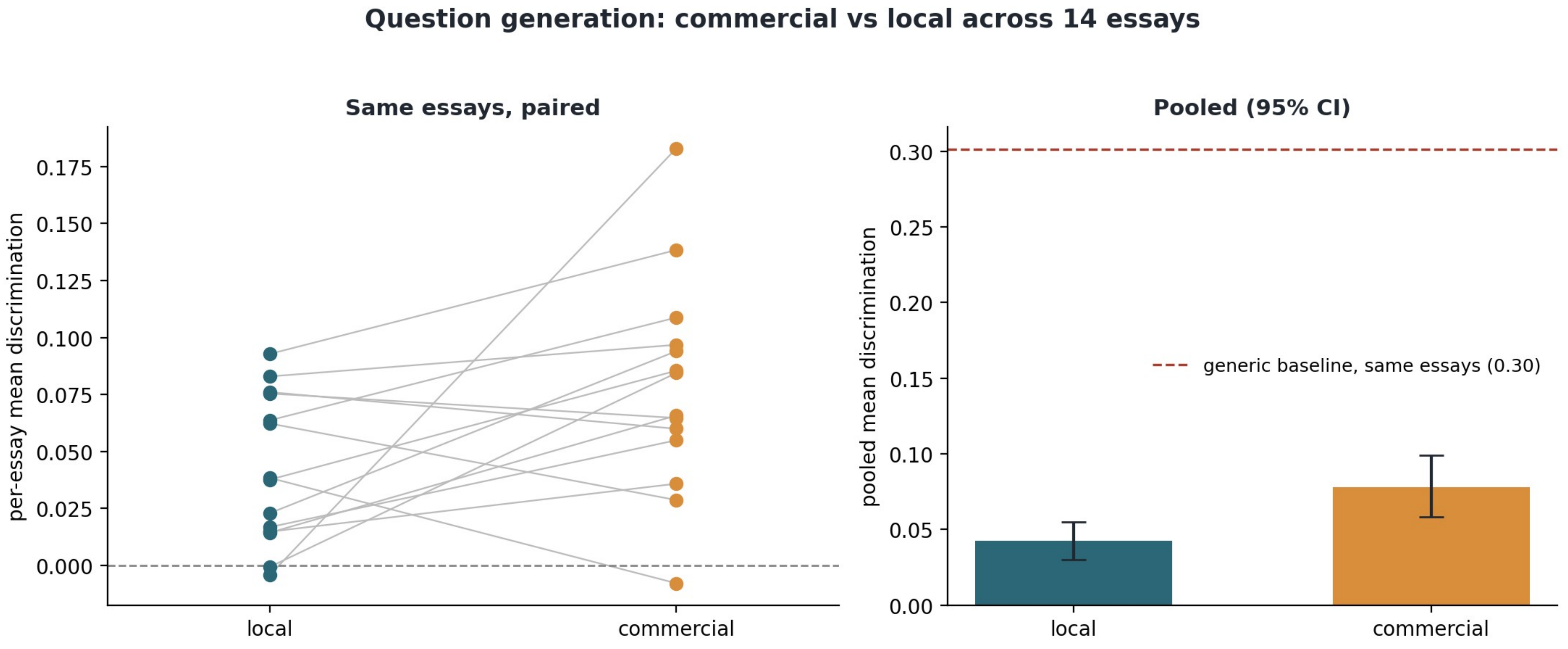
Since Meeting 4 (23 June)

- Backend A is live: a free-tier commercial API behind the same interface as the local model (ATU could not provide credits; Aoife Hill confirmed).
- The commercial-vs-local comparison ran at scale, with paired statistics, then again on a tighter fixed-claim design.
- Three more components trained: the Bloom's classifier, the claim extractor, and the output assembler producing the lecturer's PDF guide.
- Backend B executed on your sign-off: the QLoRA fine-tune ran, its first result turned out to be a metric-gaming artifact I caught by auditing the output, and a re-run on the right data fixed it. Full story in two slides.
- Writing: 60-page draft, your Chapter 1 feedback applied, 24 references all verified against the real papers, three new results sections this week.
- The 14 July presentation is updated with all of this and ready for a run-through.

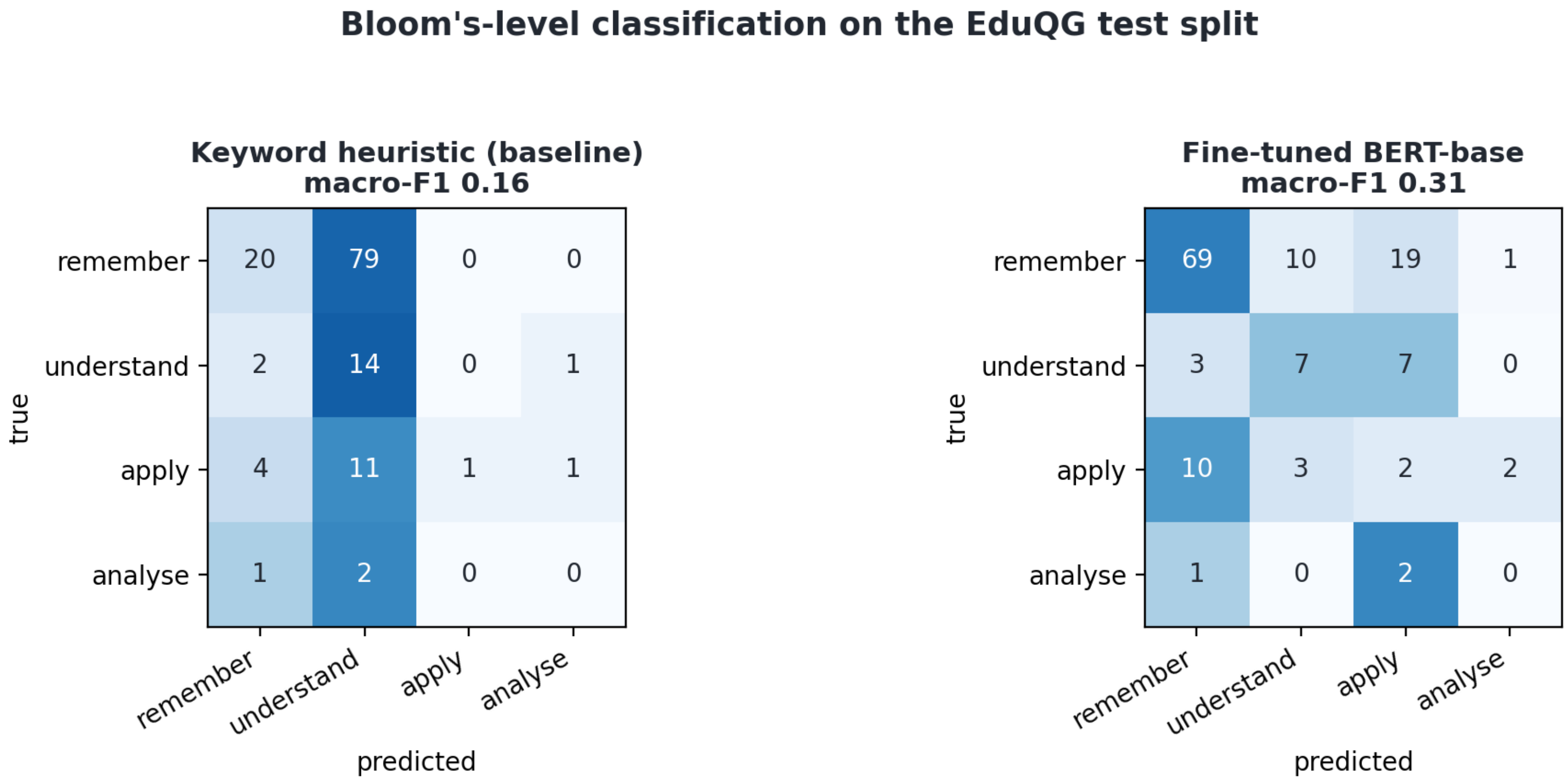
Commercial vs local, at scale and paired

Fourteen essays, 126 questions per backend, identical scoring. Local Llama 3.1 8B 0.042 [0.030, 0.055] vs free-tier Gemini 0.078 [0.058, 0.098]. Paired difference 0.036 (t p = 0.040, Wilcoxon p = 0.030): a small but significant commercial edge; the local model is competitive before any fine-tuning.



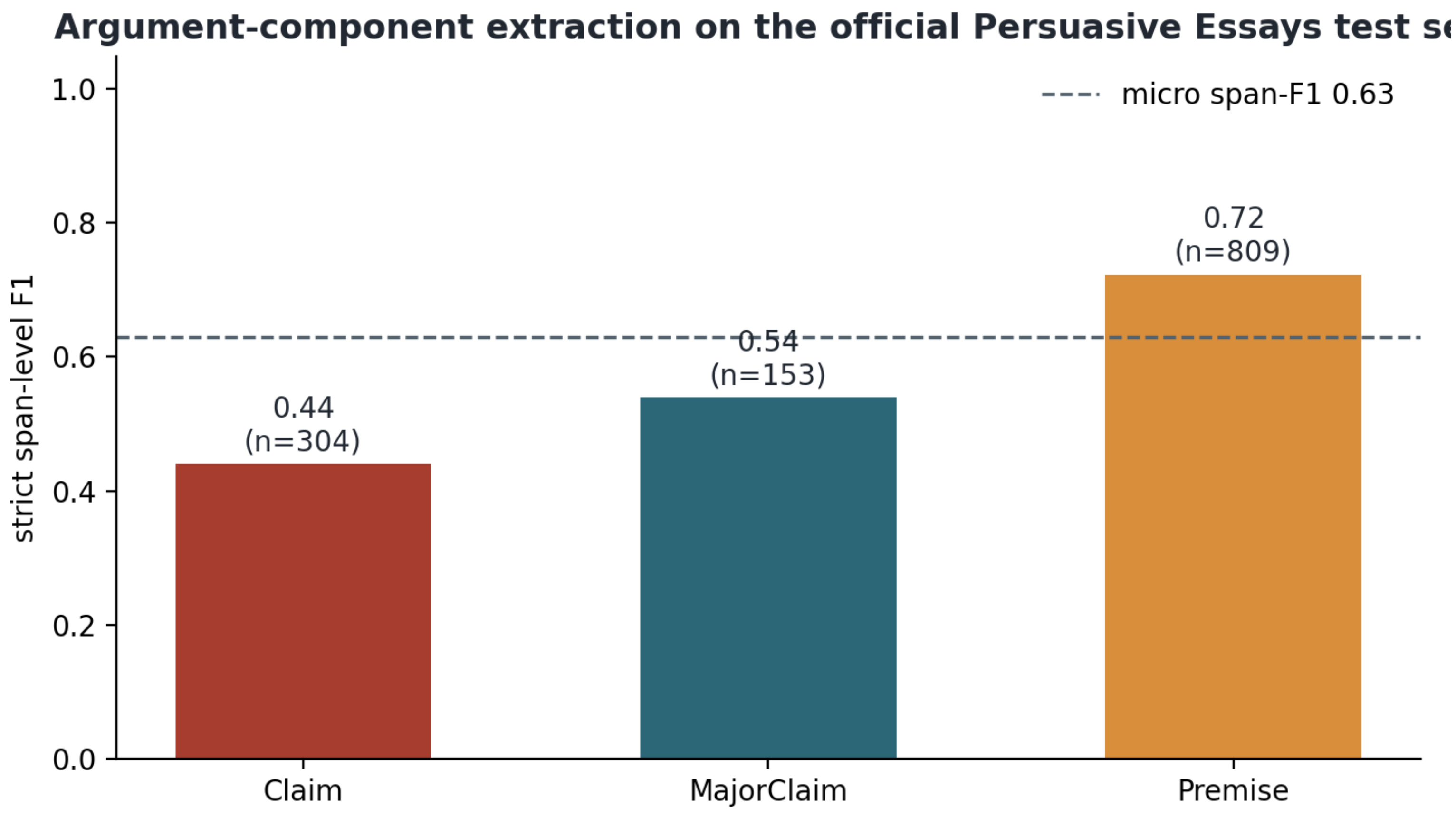
Bloom's classifier: trained, and honest about the data

BERT-base on EduQG's 903 Bloom-labelled questions vs the keyword heuristic, same test split: macro-F1 0.31 vs 0.16, accuracy 0.57 vs 0.26. The label supply is the ceiling: 73% of labels are the lowest level, and nobody learns 'analyse' from 19 examples.



Claim extractor: trained on Persuasive Essays

DeBERTa-v3-base BIO tagger on the official 322/80 split, strict span-level scoring: micro-F1 0.63 (premises 0.72, major claims 0.54, claims 0.44). Claim boundaries are the corpus's known-hard case; dedicated CRF architectures report higher, stated plainly.



The pipeline now produces its end product

The assembled Verification Interview Guide, generated end to end with live models: the trained detector's score with its limits stated plainly, the SHAP-validated drivers, claims quoted to exact sentences, questions tagged by the trained classifier, and a suggested rubric. First page states the position: evidence for a conversation, not an accusation.

Verification Interview Guide

Submission: 3108a | **Generated:** 2026-07-02 | **Question backend:** local:llama3.1:8b

> This guide is evidence for a conversation with the student, not an accusation. The detector's judgement is fallible, and the fair use of this document is to ask, listen, and weigh the answers.

1. Detection summary

The trained detector scores this submission at **0.9996** probability of being AI-generated, so it is **flagged as likely AI-generated**. No score of 1.0 exists on this scale: the model is never certain, only confident. On matched in-domain test data the detector's F1 is 0.99; on out-of-domain academic text its false-positive rate rises sharply, so treat the score as a reason to talk, never as proof.

What drives decisions like this one (validated by faithfulness testing):

- Average word length: consistently longer words push a text toward the AI class.
- Vocabulary richness: a wider, less repetitive vocabulary pushes toward the human class.
- Sentence-length variation: humans vary sentence length more; uniform sentences look AI-written.
- Auxiliary-verb and determiner densities: small grammatical habits that differ between the classes.

2. The student's claims and where they come from

Each claim below was extracted from the submission and is quoted to its exact sentence numbers, so nothing here is invented: every question can be traced to the student's own words.

Claim 1: The three texts (Dickinson's poetry, Shakespeare's Othello, and Austen's Emma) represent distinct literary genres but share certain commonalities in their exploration of universal themes and ideas.

Source in the submission:

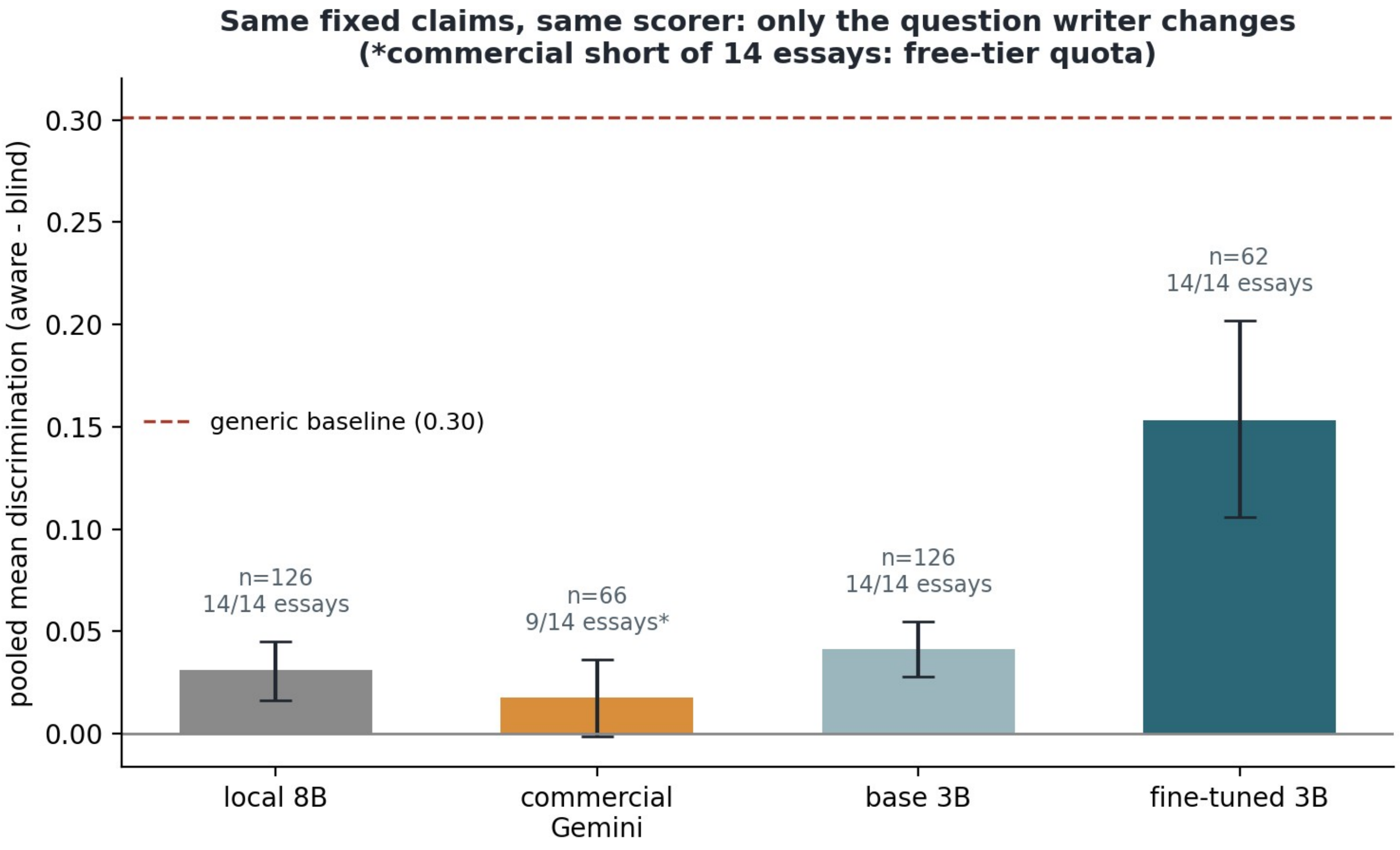
> [1] These three texts represent distinct literary genres: poetry, drama, and prose fiction respectively. [27] Despite these differences, all three texts share certain commonalities. [31] These texts represent distinct literary genres but share certain commonalities in their exploration of universal themes and ideas.

Verification questions:

- How did you determine that Dickinson's poetry represents a distinct literary genre from Shakespeare's Othello and Austen's Emma? What specific examples or features of each text led you to this conclusion? *(Bloom: understand)*

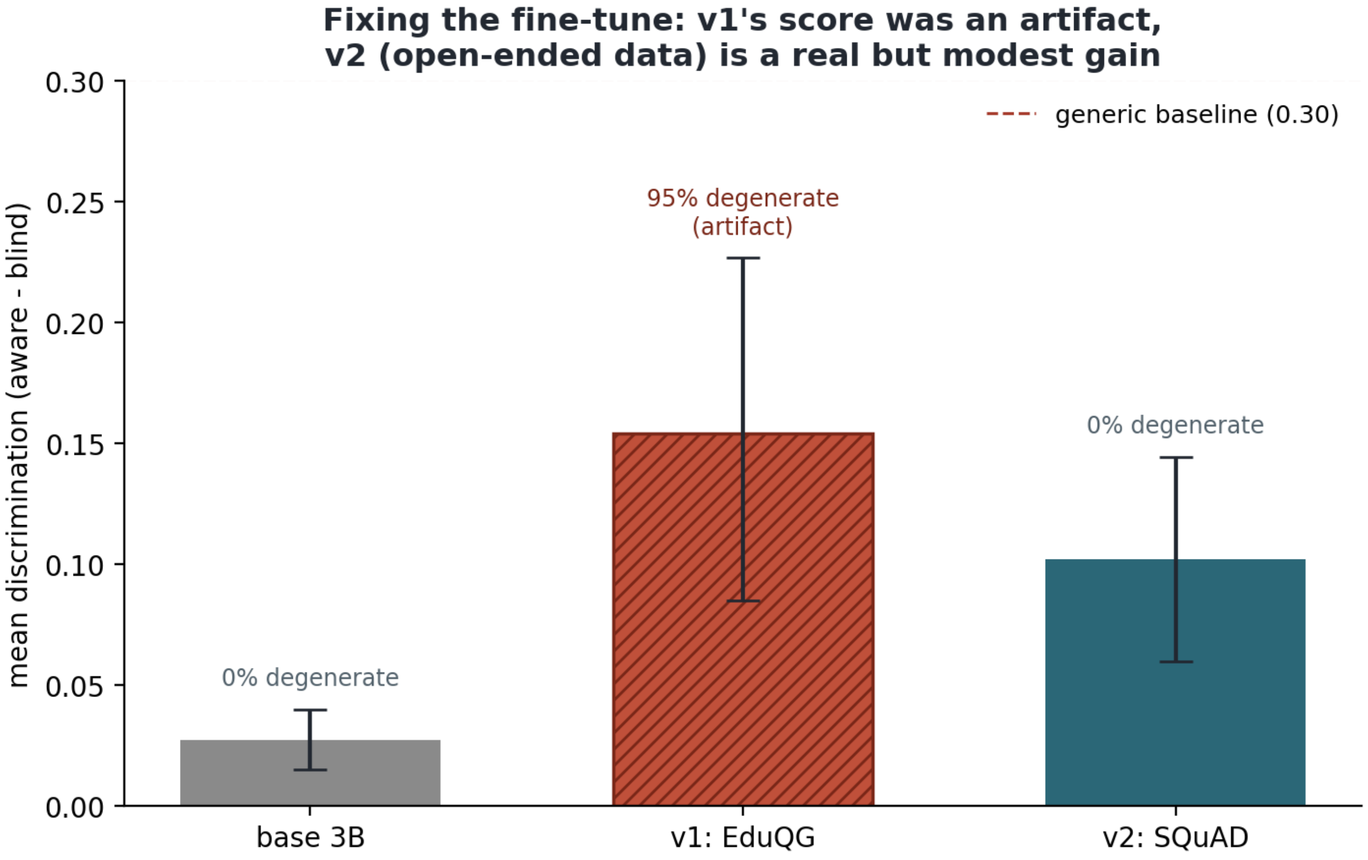
Your sign-off, used the same week: the tighter comparison

One fixed claim set per essay (neutral extractor), and every question writer answers the SAME claims, same scorer. On this tighter design the earlier commercial edge disappears (Gemini 0.017 vs Llama 8B 0.031, base 3B 0.041; paired p = 0.20), suggesting part of it came from claim selection, not question writing. The fine-tuned bar posts the top score, but the next slide is why that number cannot be taken at face value.



The week's real story: a score that was lying

The v1 fine-tune (EduQG) scored 0.154, apparently beating everything. Reading its output: 95% degenerate multiple-choice stems ('Which of the following is correct?', no options), which game the metric: that contentless stem alone scores 0.44, above every real question writer. Re-fine-tuned on open-ended data (SQuAD), everything else identical: 0% degenerate and a real gain over base (0.102 vs 0.027, p = 0.0003), still below the generic baseline. Diagnosis confirmed: data format dominates.



Writing: your Meeting 4 feedback, closed out

- Introduction: two to three opening paragraphs before the background; background elaborated; problem statement expanded.
- References: mandatory and done as I write. 24 sources, every one verified against the real paper (arXiv API, ACL Anthology, PMLR, dblp, publishers); References chapter generated from the verified list.
- Outline: now a small paragraph per chapter, as you described.
- Bullet spacing: root-caused in the document builder (lists lacked the 1.5 line spacing of body text) and fixed; verified in the rendered pages.
- Three new results sections this week: the fixed-claim comparison (8.8), the quality audit behind the fine-tune score (8.9), and the corrected fine-tune (8.10).
- Draft: 60 pages, eight chapters plus references. You have the Word version for comments.

This week's plan, and the 14 July run-through

- Backend B next step: build a reasoning-style training set (distilled from the pipeline's own verification prompts) and re-fine-tune; SQuAD proved the data format is the lever, now point it at the right format.
- Judges two and three: executing the approved capped spend on Claude and GPT for cross-model agreement (Krippendorff's alpha).
- Commercial arm refill: the fixed-claim run covered 9 of 14 essays before free-tier quota ran out; the framework resumes, so it completes on the next quota window.
- AICS: the call or email, when convenient; the paper draft is ready to align to it.
- The graded presentation is Monday 14 July: 12 slides, about 12 minutes, every number traceable, now including the fine-tune story told honestly. Could we do a 10-minute run-through today?